

Faucet Image Matching Task

Overview

You are provided with a small dataset of faucet images along with their associated metadata. Your task is to build a basic image-matching system: given a query image of a faucet, your system should return the closest matching faucet from the dataset.

This assignment evaluates your ability to work with computer vision, embeddings or similarity search, metadata usage, and design of a small end-to-end solution.

Provided Data

Dataset Link: [FaucetIdentification](#)

1. images/

A directory containing faucet images. Each image corresponds to exactly one faucet in the dataset. These are training images.

2. testing_images/

A directory containing faucet images clicked manually with a camera. These photos are intended to be used for querying the system and finding the closest match from images/ dataset. These are testing images.

3. faucet_metadata.json

A JSON file containing metadata for each faucet. Each object corresponds to an image, keyed or named such that you can map metadata to the image.

Example metadata (actual):

```
{
  "images/6323759501000_0.png": {
    "ArticleNr": "6323759501000",
    "ArticleNr_masked": "6323 759.501.000",
    "MasterArticleNr": "6323759",
    "Brand": "Laufen",
```

```
"VarDim_Color": "Verchromt",
"Farbe": [
  "Chrom"
],
"Title": "Einlochmischer Laufen Twinplus Slim Eco+, ohne
Ablaufventil, Schwenkauslauf, A 160 mm, , Verchromt",
"Description_long": "Einlochmischer Laufen Twinplus Slim Eco+,
ohne Ablaufventil, Schwenkauslauf, A 160 mm, Kaltwasser-Position des
Hebels in der Mitte, Schlauchanschlüsse ¾\"",
"Description_short": "Einlochmischer Laufen Twinplus Slim Eco+,
ohne Ablaufventil, Schwenkauslauf, A 160 mm,",
"ImageURL":
"https://profishop.sanitaastroesch.ch/multimedia/Web/PG1/06323759_501_0
00.png",
"Price": 347,
"EAN": "7612738004842",
"Warengruppe": [
  "Waschtischarmatur"
],
"Produktlinie": [
  "Twinplus Slim"
],
"Montageart": [
  "Platte/Armaturenloch"
],
"dimensions": {
  "Höhe~mm": [
    "262"
  ]
},
"TEAM-Nr": "712128",
"Availability": "True",
"VarDim_ColorCode": "501 - Verchromt"
}
}
```

Task Requirements

1. Build a faucet-matching system

Given a query image (not part of the dataset), your system should output:

- The closest matching faucet image from the dataset
- A confidence score or similarity value
- ID of the matched faucet (and optionally the metadata of the matched faucet)

2. Task Scope

You are free to choose any technical approach. Use the provided metadata however you see fit. You can use any approach you like, including LLMs, using pretrained models, training your own model, etc.

3. Optional Enhancements

Any of the following will be considered a bonus:

- Using faucet metadata in the ranking step
 - Visualizing the matching process
 - Building a minimal API or script interface
 - Improving similarity scoring with fine-tuning or weighting
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Expected Output / Deliverables

1. **A Python script or notebook containing:**
 - Dataset indexing
 - Embedding computation
 - Matching logic
 2. **Instructions on how to run your solution**
 - Any environment setup
 - Example command for querying
 3. **A brief explanation (max 1 page) describing:**
 - Your research and potential solutions.
 - The approach that you decided to use for building this solution.
 - Any tradeoffs or assumptions
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Evaluation Criteria

- **Correctness:** Does your system return reasonable matches?
 - **Clarity:** Is your code readable and structured well?
 - **Technical choices:** Choice of model, search method, metadata usage.
 - **Efficiency:** Avoiding unnecessary recomputation or heavy models.
 - **Maintainability:** Can the system be extended?
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Submission

Provide a GitHub repository or a zipped folder containing:

- Code
 - Instructions
 - Any dependencies listed in requirements.txt
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Notes

- You do not need to train a model unless you choose to.
 - Pretrained embedding models are allowed and encouraged.
 - Focus on correctness and clarity over complexity.
 - The faucet brands are German, so the metadata is provided in German.
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If anything in the task description is unclear, feel free to ask us or take your assumptions and proceed accordingly.